INTERNATIONAL MONETARY FUND

CHAD

REQUEST FOR A THREE-YEAR ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

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Chad's public domestic and external debt indicators have remained relatively stable in the last few years. However, the risk of debt distress remains high even after the decision to cancel the non-concessional Master Facility Agreement (MFA) signed with the Eximbank of China in August 2011, and the US$ 1.4 billion oil sales advance contracted in mid-2014 to finance the purchase of a 25 percent stake in the largest oil consortium operating in Chad. The projected steady decline in oil exports and associated fiscal revenues (barring new discoveries) is a critical challenge over the medium-term requiring a steady fiscal adjustment and economic diversification.

BACKGROUND

Recent Developments in Public External Debt

1. Chad's external public debt has hovered around 20 percent of GDP since the start of oil production in 2003/04. Debt owed to multilateral institutions and traditional bilateral donors has been basically constant in nominal terms over that period. Chad has instead relied more on non-traditional creditors (i.e., Libya, China) and commercial credits to address its financing needs, including to tackle its infrastructure deficit.

2. The Chadian authorities have taken steps to tackle the debt sustainability concerns identified in the 2012 Article IV DSA. In December 2013, the authorities formally communicated to Eximbank China their decision to cancel the US$2 billion (14.9 percent of 2013 GDP) Master Facility Agreement (MFA) signed in August 2011 noting that the MFA is not legally in force in Chad as it was neither formally approved by the National Assembly, nor and had disbursement taken place under the MFA.

3. The external public debt-to-GDP ratio is estimated to have risen from 19.9 percent of GDP in 2012 to 22.7 percent of GDP in 2013 mainly on account of external commercial borrowing in the form of oil sales' advances. The authorities signed two agreements in May and August of 2013 for a total of US$600 million (4.5 percent of GDP) with a commercial partner (Glencore Energy). These two loans are non-concessional and have to be reimbursed over a two-year period (against crude oil shipments).[1]

4. A new commercial borrowing operation which was contracted by SHT (a state-owned oil company) to finance the purchase of Chevron's shares in Chad's largest oil consortium in June 2014, is estimated to have elevated external public sector debt to 26.6 percent of GDP in 2014. To finance the acquisition of income-generating assets, SHT contracted a loan from Glencore Energy for a total of US$ 1.4 billion under commercial terms. The loan is to be repaid against crude oil shipments over four years (2014-18). This financing modality is similar to standard "carried equity" schemes used to finance the participation of governments as partners in oil joint ventures. Those schemes are typically not classified as "debt".[2] However, in this particular case staff sees a case for treating the operation with Glencore Energy as debt mainly because the prepayment agreement specifies that other crude oil belonging to the government could eventually be used to repay the oil sales advances in case of shortfalls.

[1] The DSA simulation assumes that the US$600 million loan is amortized over two years starting in 2014.

[2] See for example the discussion on the Decision Point document for the Republic of Congo (EBS/06/17, Appendix II).

Table 1. Chad: External Debt Stock at Year-End, 2001–2013
(Billions of CFA francs)

	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012 Est.	2013 Est.
Total	794.3	786.3	736.6	797.2	898.9	896.2	794.0	782.3	781.8	1,066.8	1,301.3	1,311.5	1,501.8
(Percent of GDP)	56.0	50.2	40.9	30.2	25.6	23.1	19.2	16.9	17.9	20.2	22.7	19.9	22.7
Multilateral	678.1	687.7	652.5	715.3	810.2	805.5	718.6	706.6	677.7	757.7	742.2	741.9	680.7
IMF	65.3	67.3	57.0	47.7	47.5	37.4	25.4	19.0	12.9	8.3	5.2	2.5	0.9
World Bank/IDA	380.6	398.3	394.0	444.5	507.8	486.1	453.4	422.0	402.7	439.5	422.3	423.3	398.2
African Development Fund/Bank	182.8	169.8	159.9	168.5	179.8	205.8	173.7	182.4	185.5	208.7	210.2	204.5	191.7
EIB	3.9	7.9	7.3	13.0	13.0	12.4	9.9	9.8	8.7	9.4	10.0	10.3	2.0
Others	45.5	44.4	34.2	41.6	62.0	63.8	56.2	73.4	67.9	91.8	94.4	101.3	87.9
Bilateral	116.2	98.6	84.1	81.9	88.8	90.7	75.4	75.7	104.1	309.1	270.6	257.3	518.9
Paris Club official debt	30.2	25.8	24.0	25.2	24.3	23.2	23.6	19.2	16.6	32.0	16.1	17.4	13.9
Non-Paris Club official debt	86.1	72.7	60.1	56.7	64.4	67.5	51.8	56.5	87.5	277.2	254.5	240.0	505.0
of which: China, People's Republic	28.6	25.4	22.0	13.6	15.4	13.9	-	3.5	20.4	125.1	43.8	24.7	23.3
Libya	-	-	-	-	-	-	-	-	-	96.2	144.2	153.1	138.2
India	-	-	-	-	-	11.3	14.8	22.7	21.4	22.9	22.3	21.3	18.9
State-Guaranteed											288.5	312.3	302.3

Sources: World Bank, Chadian authorities, selected creditors and staff estimates.

The concept of debt used here includes both contracted and guaranteed debt. The end-2013 external public debt stocks are estimated by staff based on World Bank Debt Reporting System (DRS) with recent updates for end-2012 debt stock numbers (which are still preliminary), Chad's debt management office's estimates of 2013 disbursements and amortization, and Ministry of Economy and Planning estimates of 2013 project loan disbursements. The official external debt stock data underestimate the actual level of external debt. Most notably, the debt registry does not capture the loans from China for the state's stakes in the joint-venture refinery and in the Baoré cement factory, and other government-guaranteed debt. Also, project loan disbursements are recognized only after a long lag. Both text tables have discrepancies with corresponding fiscal or balance of payments flow estimates, giving rise to residuals in the sustainability analyses.

Status of Implementation of Debt Relief Initiatives

5. The authorities are seeking to achieve debt relief under the HIPC Initiative. Chad reached the HIPC decision point in 2001, but several previous efforts to attain the completion point failed due to political instability, insufficient commitment to reforms, and weak implementation capacity, including insufficient fiscal controls. In May 2013, the authorities approved a new Poverty Reduction Strategy (PRS) covering the period 2013-15 (a joint IDA-IMF Staff Advisory Note was issued to the Executive Board in July 2013 (EBD/13/50)). In July 2013, an SMP for the period June-December 2013 was approved by Fund management, which was completed satisfactorily.[3] The successful completion of the SMP opened the way

[3] The SMP aimed at sustaining recent improvements in macroeconomic outcomes, while advancing public financial management reforms and implementing Chad's new PRS. Most quantitative targets were met, including those on the underlying fiscal policy stance (the NOPD), the level of poverty-reducing social spending, and arrears' accumulation. However, the oil sales advance operation implied that the ceilings on contracting or guaranteeing new non-concessional external debt were surpassed.

for an ECF request, which could unlock debt relief under the HIPC and MDRI Initiative, if the first review of the IMF supported program is completed and the authorities comply with the HIPC floating completion point conditions.[4]

6. Realizing debt relief under the Enhanced HIPC Initiative and the Multilateral Debt Relief Initiative (MDRI) would help significantly reduce external debt and provide important fiscal space to address development needs. Debt relief under the MDRI would cover the full stock of MDRI-eligible debt owed to IDA and the African Development Fund that is outstanding at the date that the Completion Point is reached. In nominal terms, this could total about US$850 million and would imply a reduction in debt service of about US$40 million per year over the next 20 years. A scenario which assumes potential debt relief from multilaterals only (about US$850 million)[5] would result in a moderate risk of debt distress, but the PV of the debt-to-export ratios would still be close to its threshold toward the end of the projection period in the baseline scenario. This scenario is preliminary and a detailed debt reconciliation exercise based on most recent debt data will need to be conducted to firm up these estimates and include potential debt relief from other creditors before reaching the HIPC completion point.

Recent Developments in Public Domestic Debt

7. The stock of public domestic debt has grown, reaching 7.7 percent of GDP at end-2012 (and an estimated 8.3 percent of GDP in 2013). This growth was the result of drawing on central bank statutory advances, the sale of over CFAF 100 billion of five-year savings bonds (in 2011), and a CFAF 50 billion loan from Republic of Congo (in 2012). The stock of public domestic debt includes a balance of CFAF 50.3 billion in treasury arrears. In 2013, the authorities placed a five year savings bonds for CFAF 90.2 billion in the CEMAC regional market with a 6 percent coupon and contracted a CFAF 15 billion loan from Equatorial Guinea. Treasury arrears have declined to CFAF 40 billion at end-December 2013 from CFAF 50.3 billion at end-December 2012.

[4] The World Bank and IMF have been closely working with the authorities on assessing compliance with the HIPC floating completion point conditions.

[5] These estimates are preliminary. No detailed information is available for the other creditors that would also potentially provide debt relief.

Table 2. Chad: Stock of Domestic Debt at Year-End, 2006-2013

	2006	2007	2008	2009	2010	2011	2012	2013
Total	122.3	123.1	142.7	238.6	286.9	464.1	483.6	550.5
(Percent of GDP)	3.2	3.0	3.1	5.5	5.4	8.1	7.7	8.3
Central Bank statutory advances	-	17.0	21.6	141.6	208.6	208.6	187.7	166.9
Rescheduled debts	71.8	48.6	56.8	58.2	67.9	52.9	44.8	43.7
Treasury arrears	24.8	26.1	41.1	25.7	3.1	56.4	50.3	39.9
Legal commitments	13.2	12.5	10.8	10.1	4.6	21.2	49.8	70.5
Standing payment orders	11.5	18.1	11.5	2.1	1.8	3.1		
National Savings Bond	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
2011 Savings Bond	-	-	-	-	-	121.0	100.0	83.3
Republic of Congo	-	-	-	-	-	-	50.0	40.0
Equatorial Guinea	-	-	-	-	-	-	-	15.0
2013 Savings Bond	-	-	-	-	-	-	-	90.3

Source: Chadian authorities.

Debt Burden Thresholds under the Debt Sustainability Framework

8. For the purpose of defining debt burden thresholds under the Debt Sustainability Framework (DSF), Chad is classified as a weak policy performer. Chad's rating on the World Bank's Country Policy and Institutional Assessment (CPIA) improved to 2.6 in CY2013 (on a scale from 1 to 6) reflecting significant progress in governance during 2013.

Table 3. External Public Debt Burden Thresholds for "Weak Policy Performers" Under the Debt Sustainability Framework

Present value of external debt in percent of:	
GDP	30
Exports	100
Revenue	200
External debt service in percent of:	
Exports	15
Revenue	18

Source: 2011 IDA Country Performance ratings (methodology and results).

DSA ASSUMPTIONS

9. The assumptions underpinning the DSA are in line with those under the previous DSA and reflect the newly adopted policy of a uniform discount rate of 5 percent, the cancellation of the US$2 billion non concessional MFA, and the contracting of two non concessional oil sales' advance operations (US$600 million in 2013 to cover budget shortfalls and US$ 1.4 billion contracted by the State in June 2014 to finance the purchase of income-generating equity shares in the largest oil

consortium operating in Chad). The DSA analysis is conducted over a baseline (current policies) scenario that reflects the stated intentions of the authorities to pursue fiscal consolidation, while protecting priority spending and investing adequately in physical infrastructure to prepare the country for the post-oil era. The financing needs in 2014 are small compared to other years (amid the high exceptional oil related receipts linked to the settlement of a legal dispute) and are mostly covered through multilateral borrowing from the IMF, the African Development Bank and the Islamic Development Bank (70 percent). Over the medium to long term it is assumed that multilateral and bilateral borrowing stabilize at 55 and 45 percent respectively. Due to the new commercial loan the grant element drops in 2014 before increasing again in 2015 and stabilizing over the medium and long run to about 35 percent.

10. Oil production and revenue: Chad's medium- and long-term macroeconomic outlook is characterized by a temporary sharp increase in oil production in 2014–16 and a steady decline of oil production over the subsequent sixteen-year projection period (Box 1).[6]

- Oil production is expected to rise from 100,000 bpd in 2013 to about 225,000 bpd in 2016-17. This second oil boom will likely be temporary; proven reserves in the new fields are much smaller than those in the original Doba basin and will also likely be nearly exhausted around 2030. Hence, oil production and exports are projected to decline steadily to negligible levels beyond 2030. Obviously, these prospects might change with new oil exploration activities.

- Chad's oil trades below the WEO reference price, reflecting a quality discount and transport[7] cost. For the medium term (five-year horizon) the price of Chadian oil is assumed to drop from US$103.9 per barrel (all discounts included) in 2013 to about US$87.8 per barrel in 2017–18, in line with the trend projected in the IMF's World Economic Outlook (WEO). Over the long term, the price is assumed to increase by 2 percent per year in U.S. dollar terms (consistent with the assumption used by the IMF in long-term projections for other sub-Saharan African countries).

11. Fiscal policy: The analysis assumes a sustained fiscal adjustment throughout the entire projection in transition to the post-oil era, with the non-oil primary deficit (NOPD) reduced from about 16 percent of non-oil GDP in 2014 to 14-15 percent of non-oil GDP by 2018. The increase in oil production over the medium-term is expected to push oil revenues above 18 percent of non-oil GDP in 2016-17. Over the

[6] Oil production at the Doba oilfield (exploited by the Esso-led consortium) started in 2003, reached its peak of 63 million barrels in 2005 and, absent other oil developments, will decline with annual output projected to become negligible beyond 2030. In 2011, oil production began at a second oil field, Bongor, operated by CNPC (about one-third the size of the Doba field). At present, Bongor produces crude oil at a rate of about 15,000 barrels per day as feedstock for the oil refinery, resulting in a near complete substitution of petroleum product imports. Exports of crude oil from Bongor are expected to start in 2014 and reach 105,000 bpd in 2017. Oil from another smaller operator (Caracal, formerly Griffiths) started to be produced in late 2013 and are expected to peak at 60,000 bpd in 2016.

[7] The discount ranged from US$2.6 to US$7.2 in 2013, partly reflecting developments in global energy markets with the increased importance of shale gas and specifically oil exports from Angola, Saudi Arabia and South Sudan. Future quality discounts will also reflect the quality of oil from the new shippers (CNPC and Caracal).

longer term, it is assumed that dwindling oil revenues will be partly offset by a decline of total government spending from 30.3 percent of non-oil GDP in 2013 to 23 percent in 2033-34, while the NOPD will be adjusted gradually to reach 0.8 percent of non-oil GDP by 2033. The latter is projected to be achieved mainly by: (i) increasing non-oil revenues (from about 9 percent of non-oil GDP at present to about 18 percent of non-oil GDP by 2034); (ii) reducing total investment outlays from over 14 percent of non-oil GDP to below 10 percent of non-oil GDP; and (iii) cutting recurrent spending, notably, by streamlining transfers and subsidies to public enterprises (at present, jointly accounting for about 5.5 percent of non-oil GDP).

Box 1. Macroeconomic Assumptions, 2013–2033

Baseline Scenario:

Real GDP growth is driven by a sharp growth of oil production in the next four years and a steady decline in oil production over the following seventeen years with upside risk over the short and long run given current and future oil exploration activities. A moderate scaling-up of foreign-financed investments (see below) is assumed to raise non-oil GDP growth in the medium term (5.2 percent in 2014–18). Beyond that, non-oil GDP growth is assumed to stabilize at 4.2 percent in the long term. Inflation is assumed to stabilize at 3 percent, consistent with the CEMAC convergence criterion. The external current account will remain in deficit throughout the forecasting period although it will shrink rapidly in 2014–17, with the increase in oil exports. With the decline in oil production over the long-term, the current account deficit is expected to increase due to the drop in exports however the deficit is contained since oil related imports are decreasing as well.

The fiscal outlook features a temporary increase in oil revenues in the medium term but a decline in the long term. The authorities are assumed to reduce the NOPD to 13 percent of non-oil GDP by 2018. In the medium and long term, the authorities adjust by increasing non-oil revenue effort, streamlining exceptional security spending and subsidies to public enterprises, while maintaining an appropriately high level of investment and social spending to ensure steady growth and poverty reduction (the policy actions underlying these structural changes are listed in the SR, section C).

External financing: Project loans are assumed to pick up from about 1 percent of non-oil GDP in 2013-14 to 2.5 percent of non-oil GDP over 2015-19 and then fall to 2.1 percent of non-oil GDP in the long term. Project grants will increase from about 2.5 percent of non-oil GDP in 2013-14 to 2.6 percent of non-oil GDP over 2015-17 and 2.7 percent of non-oil GDP for the remainder of the forecasting period. This is assumed based on historical averages, with 60 percent grants and 40 percent in loans from official multilateral and bilateral creditors. The baseline does not take into account HIPC and MDRI debt relief for which Chad is eligible.

The domestic financing assumptions include reimbursement of BEAC statutory advances by 2015, no accrual of domestic arrears, and limited issuance of domestic debt instruments from 2015 onward.

Text Table 1. Chad: Medium-Term Projections

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2014–17 Avg.	2018–34 Avg.
Real GDP growth (percent per year)	0.1	8.9	3.9	9.6	6.7	9.7	6.9	3.1	3.3	8.2	2.9
Oil	-0.3	-4.0	-6.8	30.3	15.8	32.0	14.2	-3.1	-2.8	23.1	-5.8
Non-oil	0.2	11.6	5.9	6.3	4.9	4.9	4.9	4.9	5.0	5.3	4.2
Consumer price inflation (percent per year)	1.9	7.7	0.2	2.8	3.1	3.0	3.0	3.0	3.0	3.0	3.0
External current account balance (percent of GDP)	-5.6	-8.7	-9.5	-7.2	-7.1	-5.9	-4.2	-4.5	-4.7	-7.1	-5.6
					(In percent of non-oil GDP)						
Government revenue and grants	34.6	35.0	26.8	28.8	26.7	30.2	31.5	33.3	31.6	29.3	25.7
Of which: oil revenue	24.8	23.2	15.5	17.3	14.9	17.8	18.7	19.9	17.7	17.2	8.6
Of which: grants	2.2	3.8	2.3	2.7	2.6	2.6	2.6	2.6	2.6	2.6	2.7
Government expenditure (commitment basis)	31.3	34.4	30.3	29.3	29.0	30.0	29.9	31.1	30.7	29.5	26.5
Overall fiscal balance (incl. grants; cash basis)	0.6	2.2	-6.4	-0.5	-2.5	0.3	1.7	2.9	1.1	-0.3	-0.5
Non-oil primary fiscal bal. (excl. grants; commitment basis)	-19.6	-20.1	-17.5	-15.9	-14.1	-14.6	-14.0	-14.8	-13.9	-14.7	-7.1
Memorandum items:											
Chadian crude oil price (US$/barrel)	97.7	102.0	103.9	103.0	97.9	93.0	88.6	86.9	85.5	95.6	97.2

Sources: Chadian authorities; and IMF staff estimates and projections.

EXTERNAL DEBT DSA

Baseline (Current Policies) Scenario

12. The evolution of external debt beyond 2014 is driven by a reasonable volume of project and budget support loans from both traditional and non-traditional donors. There is no external private debt in Chad. Apart from the newly contracted loan in the form of an oil sales advance new borrowing is projected to be mainly on concessional terms from multilaterals (70 percent)[8], Paris Club official (20 percent) and non-Paris Club official (10 percent), in the medium to long run it is assumed that Chad borrows 55 percent from multilaterals, 35 percent from the Paris Club and 10 percent from non-Paris Club officials. Due to the new commercial loan, the average concessionality of new borrowing (assuming a discount rate of 5 percent) drops in 2014 to 0.4 percent but returns to an average of 35 percent over the remainder of the estimation period (Figure 1a).

13. The comparison of the current baseline scenario with the baseline featured in the 2013 DSA reveals some differences reflecting the new commercial borrowing in the form of an oil sales advance of US$ 1.4 billion, increases in Chad's oil production projections, and lower financing needs. On the one hand, the new loan pushes external debt temporarily up in 2014 to 26.6 percent of GDP while the extraordinary oil-related receipts linked to the settlement of disputes with oil companies reduces the government's borrowing needs in 2014. On the other hand, Chad's participation in the Esso oil consortium leads to higher cash inflows and thus lowers financing needs in the long run and based on more recent information from the companies exploiting the newer oil fields, oil production projections

[8] Amid the high revenues from the environmental fine, the financing needs in 2014 are minimal and can be mostly covered through assumed Fund disbursements under the ECF and already approved loans from the African Development Bank and the Islamic Development Bank. For these reasons, the share of multilateral borrowing in 2014 is higher than in other years.

have been revised upwards. This in turn has led to higher oil exports and fiscal oil revenue projections over the medium and long term. For these reasons, debt accelerates less fast and apart from the debt service-to-revenue ratio which is only breached in 2015 (amid the new oil sales advance loan operation), debt-to-revenue and export ratios are much lower compared with the last DSA. Moreover, in the last DSA the most extreme shock (which was a drop in export growth in 2014–15) resulted in overly high debt-to-export ratios due to a high standard deviation over the past 10 years (linked to sky-rocketing growth in oil exports in 2003-04 when Chad started to produce oil). To adjust for this structural break, the standard deviation of exports is now estimated only for 2005-12. While a drop in exports is still the most extreme shock, the debt-to-export ratios over the long run is more reasonable now.

14. Under the baseline scenario, all but one of the external indebtedness indicators remain below their critical thresholds, indicating a high risk of external debt distress. The present value of external public and publicly-guaranteed debt falls below 15 percent of GDP starting in 2017 and remains well below the 30 percent of GDP threshold throughout the end of the projection horizon (Figure 1b). With the projected steady decline in oil exports beyond the 2013–17 horizon, the PV of debt is expected to rise relative to exports, but does not exceed the critical threshold and remains within the 10 percent band below the threshold till the end of the projections. Since the debt service-to-revenue ratio is breached under the baseline scenario and the ratio in 2015 is above the 10 percent band, Chad is assessed to currently have a high risk of debt distress. However, assuming the debt profile does not change, the debt rating would decline to a borderline moderate/high case after 2015.

Stress Tests

15. Under the shock scenario that assumes that key macroeconomic variables remain at their ten-year average values, the debt burden indicators, except for the debt service-to-revenue ratio, do not breach the sustainability thresholds. However, Chad's external debt burden indicators are highly sensitive to a negative shock to exports (e.g., an oil price shock). Across all indicative debt burden thresholds, the most extreme shock is a drop in export growth in 2014–15 (i.e., a one-standard-deviation drop in exports over the historical average): debt would be on a path breaching all indicative debt burden thresholds. This underscores the need for greater diversification of the economy.

16. The result of an additional shock scenario which assumes a permanent decline in oil prices, supports the high risk of external debt distress assessment. If oil prices were to fall persistently by 10 percent, two external indebtedness indicators (debt-to-exports and debt service-to-export) will breach their respective threshold under the baseline scenario. Whereas the PV of debt-to-exports is calculated to cross the threshold in 2030 and passing the 10 percent band by 2032, the critical threshold for the debt service-to-export ratio is only breached at the beginning of 2032.

PUBLIC DEBT DSA

17. The inclusion of domestic debt in the analysis does not fundamentally alter the assessment of Chad's debt sustainability. Given the moderate size of Chad's domestic debt, the baseline adjustment in the non-oil primary balance, and expected oil revenues, the public debt indicators are driven mainly by the external debt component. The domestic debt component would fall from 8.3 percent of GDP in 2013 to

3.6 percent of GDP in 2018 but will continuously rise thereafter until it reaches about 4.7percent of GDP 2033. Altogether, the public debt stock decreases to less than20 percent of GDP in 2018 before rising continuously until it stabilizes around 22 percent of GDP in 2031-33 which is 11 percentage points of GDP lower compared to the DSA under the 2013 Article IV consultation.

18. Despite this relatively low debt stock, standard stress tests indicate significant sustainability risks. In particular, reflecting the narrow base of an economy highly dependent on volatile oil and agriculture sectors, a temporary shock to real GDP growth in 2014-2015 would substantially impair public debt sustainability (Most Extreme Shock in Figure 2 and Bound Test B1 in Table 2a).

THE AUTHORITIES' VIEWS

19. The authorities continue to believe that improved oil revenue prospects and debt relief through the HIPC process would substantially dampen debt sustainability concerns. The authorities acknowledged potential concerns on debt sustainability by the recent decision to cancel the MFA with Eximbank China as a signal of the importance that they attach to this issue. On the newly contracted external loan for the 25 percent stake in the oil consortium, the Chadian government stresses that it supported the acquisition of an income-generating asset with a positive overall financial return and that it should not be classified as "debt" as it is equivalent to the standard "carried equity" schemes applied in many developing (and liquidity constrained) countries to facilitate their participation in oil joint ventures. They acknowledged the upward revision of oil revenue projections, but they are confident of further increases on the basis of new exploration rounds to be conducted following a geological mapping exercise. At the same time, they emphasized the need for debt relief to create fiscal space for additional development spending (including foreign-financed). Finally, they highlighted that strengthened economic management should help allay debt sustainability concerns and the resultant expected improvement of the CPIA score would lead to an increase of external debt burden thresholds.

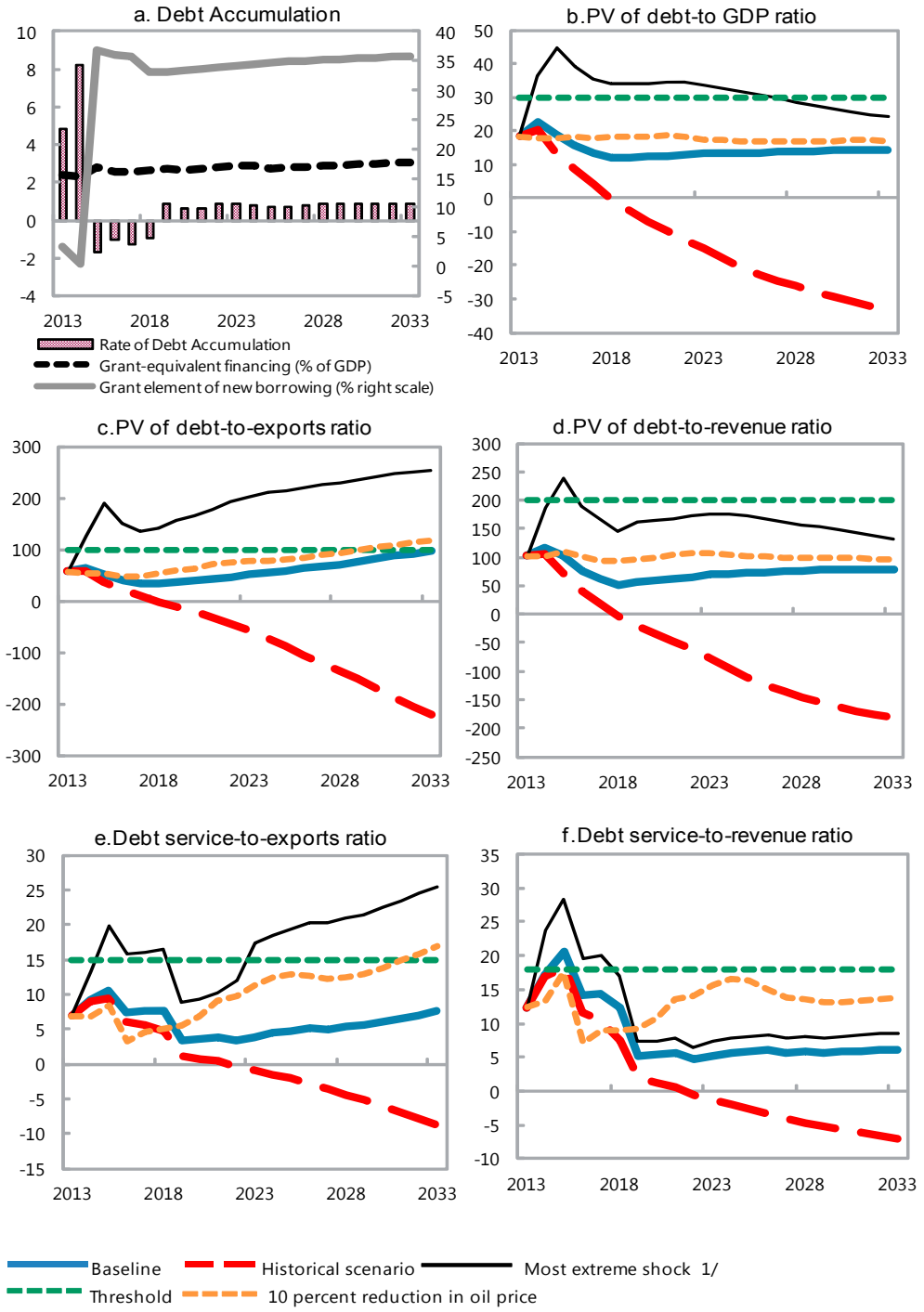
DEBT DISTRESS CLASSIFICATION AND CONCLUSIONS

20. In staff's assessment, despite improved oil production and revenue prospects in the next few years, Chad faces a high risk of debt distress. This is due to the steady decline in oil exports and associated fiscal revenues over the long term. While the government's decision to cancel the US$2 billion non concessional MFA with the Eximbank of China significantly improves the debt distress rating compared to the 2012 Article IV DSA, Chad still remains at high risk of debt distress despite a sustained fiscal adjustment unless it adopts a prudent external borrowing policy. Sustained fiscal adjustment is a necessary but not sufficient condition for minimizing the risk of debt distress. Given the exhaustibility and volatility of oil revenues, fiscal adjustment needs to be complemented by a prudent external borrowing policy and further progress in diversifying the economy. Obtaining debt relief under the HIPC initiative would reduce the debt distress to moderate, but borrowing policies would still need to be prudent.

21. Progress toward the HIPC completion point (including improvements in the CPIA score) would substantially reduce Chad's debt vulnerabilities. In addition to continued pursuit of fiscal consolidation and a prudent external borrowing policy, institutional and governance reforms should pave the way for debt relief under the HIPC process and increased access to concessional borrowing sources, thereby reducing the country's debt vulnerabilities.

Baseline Scenario

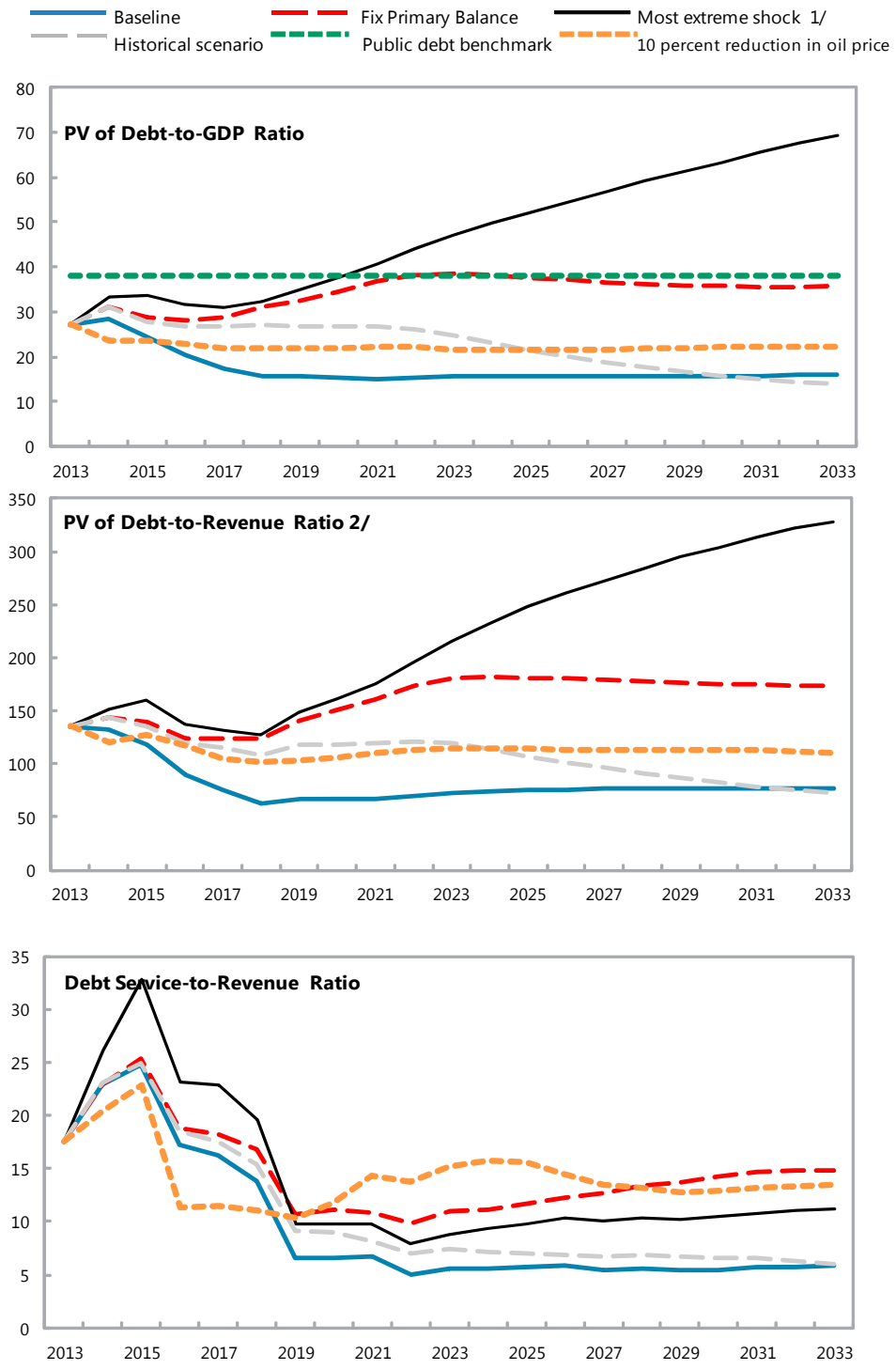
Figure 1. Chad: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2023. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Chad: Indicators of Public Debt Under Alternative Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2023.

2/ Revenues are defined inclusive of grants.

Table 1a. Chad: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010-2033

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	25.6	29.1	27.4			31.0	31.9	28.0	24.0	21.2	19.7		20.5	22.0	
of which: foreign-currency denominated	20.1	21.4	19.9			22.7	26.2	22.6	19.5	17.2	16.1		18.4	20.7	1.3
Change in public sector debt	1.7	3.5	-1.7			3.7	0.9	-3.9	-4.0	-2.8	-1.4		0.3	0.1	
Identified debt-creating flows	1.5	-4.4	-4.2			0.5	-8.2	-4.9	-6.2	-5.4	-4.3		0.2	-0.3	
Primary deficit	4.0	-2.8	-1.4	0.7	4.2	1.1	-2.2	-1.5	-3.5	-4.0	-4.1	-2.4	0.7	0.3	0.4
Revenue and grants	20.2	24.8	23.4			20.1	21.6	20.7	22.6	23.2	25.1		21.4	20.7	
of which: grants	1.3	1.6	2.5			2.2	2.2	2.0	1.9	1.9	2.0		2.2	2.4	
Primary (noninterest) expenditure	24.3	22.0	22.0			21.2	19.4	19.2	19.1	19.2	21.0		22.1	21.0	
Automatic debt dynamics	-2.4	-1.6	-2.7			-0.1	-3.0	-1.8	-2.5	-1.2	0.0		-0.4	-0.6	
Contribution from interest rate/growth differential	-4.7	-0.7	-3.5			0.6	-2.2	-1.0	-2.0	-0.7	0.5		-0.4	-0.6	
of which: contribution from average real interest rate	-1.8	-0.7	-1.1			1.6	0.5	1.0	0.5	0.8	1.1		0.2	0.0	
of which: contribution from real GDP growth	-2.9	0.0	-2.4			-1.0	-2.7	-2.0	-2.5	-1.6	-0.6		-0.5	-0.7	
Contribution from real exchange rate depreciation	2.3	-0.9	0.8			-0.7	-0.8	-0.8	-0.5	-0.5	-0.5		...	...	
Other identified debt-creating flows	-0.2	0.0	0.0			-0.5	-3.0	-1.5	-0.2	-0.2	-0.2		-0.2	0.0	
Privatization receipts (negative)	-0.2	0.0	0.0			0.0	-2.6	-1.3	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			-0.5	-0.4	-0.2	-0.2	-0.2	-0.2		-0.2	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.2	7.9	2.5			3.1	9.0	1.0	2.2	2.6	2.9		0.1	0.4	
Other Sustainability Indicators															
PV of public sector debt	...	...	21.9			27.2	28.6	24.5	20.4	17.4	15.7		15.5	15.8	
of which: foreign-currency denominated	...	...	14.5			18.8	22.9	19.1	15.9	13.4	12.0		13.3	14.5	
of which: external	...	...	14.5			18.8	22.9	19.1	15.9	13.4	12.0		13.3	14.5	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.6	-1.9	0.8			4.6	2.8	3.6	0.4	-0.2	-0.6		1.9	1.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	93.6			135.3	132.1	118.2	90.3	75.2	62.6		72.3	76.6	
PV of public sector debt-to-revenue ratio (in percent)	...	...	105.0			151.8	147.3	131.1	98.6	81.9	67.9		80.8	86.5	
of which: external 3/	...	...	69.4			105.3	117.9	102.4	76.8	63.1	52.1		69.4	79.3	
Debt service-to-revenue and grants ratio (in percent) 4/	2.9	3.6	9.5			17.6	22.9	24.8	17.2	16.3	13.8		5.5	5.8	
Debt service-to-revenue ratio (in percent) 4/	3.1	3.9	10.7			19.7	25.6	27.5	18.8	17.8	15.0		6.2	6.6	
Primary deficit that stabilizes the debt-to-GDP ratio	2.3	-6.3	0.3			-2.5	-3.1	2.4	0.5	-1.1	-2.6		0.4	0.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	13.5	0.1	8.9	9.0	10.0	3.9	9.6	6.7	9.7	6.9	3.1	6.7	2.6	3.1	2.9
Average nominal interest rate on forex debt (in percent)	1.0	2.2	3.4	1.4	0.8	3.3	4.1	3.7	3.3	2.9	2.4	3.3	2.3	2.0	2.2
Average real interest rate on domestic debt (in percent)	...	...	-0.7	-0.7	...	7.1	0.0	3.3	2.3	4.1	5.8	3.8	3.1	3.1	3.2
Real exchange rate depreciation (in percent, + indicates depreciation	15.9	-4.4	4.3	-4.8	14.4	-3.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	6.4	8.6	5.4	6.6	8.3	-3.2	4.5	2.1	2.5	0.6	-1.1	0.9	1.8	2.0	1.8
Growth of real primary spending (deflated by GDP deflator, in percer	0.1	-0.1	0.1	0.1	0.2	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	3.4	0.4	36.8	35.9	35.8	33.0	24.2	34.1	35.7	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2a.Chad: Sensitivity Analysis for Key Indicators of Public Debt 2013-2033

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	27	29	24	20	17	16	15	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	27	31	28	27	27	27	25	14
A2. Primary balance is unchanged from 2013	27	31	29	28	29	31	39	36
A3. Permanently lower GDP growth 1/	27	29	26	24	22	23	40	104
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	27	33	34	32	31	32	47	69
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	27	34	35	30	26	25	24	23
B3. Combination of B1-B2 using one half standard deviation shocks	27	34	34	30	28	27	33	42
B4. One-time 30 percent real depreciation in 2014	27	36	31	26	22	20	19	19
B5. 10 percent of GDP increase in other debt-creating flows in 2014	27	36	32	27	24	22	22	21
PV of Debt-to-Revenue Ratio 2/								
Baseline	135	132	118	90	75	63	72	77
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	135	143	135	119	115	109	120	72
A2. Primary balance is unchanged from 2013	135	144	140	124	124	123	180	173
A3. Permanently lower GDP growth 1/	135	136	127	103	95	89	181	473
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	135	152	160	137	131	127	216	329
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	135	158	168	131	113	98	113	111
B3. Combination of B1-B2 using one half standard deviation shocks	135	157	162	132	119	108	155	202
B4. One-time 30 percent real depreciation in 2014	135	166	149	114	95	80	91	93
B5. 10 percent of GDP increase in other debt-creating flows in 2014	135	168	154	119	102	88	101	101
Debt Service-to-Revenue Ratio 2/								
Baseline	18	23	25	17	16	14	6	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	18	23	25	18	18	15	7	6
A2. Primary balance is unchanged from 2013	18	23	25	19	18	17	11	15
A3. Permanently lower GDP growth 1/	18	23	26	18	18	16	11	30
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	18	25	29	22	21	19	13	24
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	18	23	26	21	20	15	8	9
B3. Combination of B1-B2 using one half standard deviation shocks	18	24	27	21	20	17	10	15
B4. One-time 30 percent real depreciation in 2014	18	26	33	23	23	20	9	11
B5. 10 percent of GDP increase in other debt-creating flows in 2014	18	23	26	21	17	15	8	8

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

Table 3a .Chad: External Debt Sustainability Framework, Baseline Scenario, 2010-2033 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**20.1**	**20.3**	**20.3**			**22.7**	**26.4**	**22.8**	**19.6**	**17.3**	**16.2**		**18.5**	**20.9**	
of which: public and publicly guaranteed (PPG)	20.1	20.3	20.3			22.7	26.4	22.8	19.6	17.3	16.2		18.5	20.9	
Change in external debt	0.7	0.2	0.1			2.4	3.7	-3.6	-3.2	-2.3	-1.1		0.5	0.2	
Identified net debt-creating flows	1.3	-1.3	2.7			4.8	-1.1	-0.6	-1.8	-0.9	1.1		3.8	5.4	
Non-interest current account deficit	**8.8**	**5.2**	**7.7**	**7.0**	**14.7**	**8.8**	**6.4**	**6.2**	**5.2**	**3.7**	**4.1**		**4.7**	**6.0**	5.2
Deficit in balance of goods and services	11.3	7.3	10.2			10.0	9.6	7.6	4.4	2.5	3.1		5.7	8.1	
Exports	37.9	40.7	36.8			32.4	36.1	36.6	40.2	40.5	37.3		25.9	14.9	
Imports	49.1	48.0	47.0			42.4	45.6	44.1	44.6	43.0	40.4		31.6	23.0	
Net current transfers (negative = inflow)	-5.6	-4.3	-4.2	-4.9	1.5	-4.9	-7.1	-5.4	-3.9	-3.7	-3.9		-3.9	-3.2	-3.6
of which: official	-0.4	-0.3	-0.9			-1.5	-4.1	-2.5	-1.3	-1.2	-1.3		-1.3	-1.3	
Other current account flows (negative = net inflow)	3.2	2.2	1.6			3.8	4.0	4.1	4.7	4.9	4.9		2.8	1.2	
Net FDI (negative = inflow)	**-5.2**	**-4.5**	**-4.5**	**-9.6**	**9.3**	**-3.9**	**-6.4**	**-6.1**	**-5.7**	**-3.8**	**-2.9**		**-0.9**	**-0.4**	-0.9
Endogenous debt dynamics 2/	**-2.4**	**-2.1**	**-0.5**			**-0.1**	**-1.1**	**-0.7**	**-1.3**	**-0.7**	**-0.1**		**0.0**	**-0.2**	
Contribution from nominal interest rate	0.2	0.4	0.6			0.7	0.8	0.9	0.7	0.5	0.4		0.4	0.4	
Contribution from real GDP growth	-2.3	0.0	-1.7			-0.8	-1.9	-1.6	-1.9	-1.2	-0.5		-0.4	-0.6	
Contribution from price and exchange rate changes	-0.3	-2.4	0.5			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-0.5**	**1.5**	**-2.6**			**-2.5**	**4.8**	**-3.0**	**-1.4**	**-1.5**	**-2.2**		**-3.3**	**-5.3**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	14.5			18.3	22.7	19.0	15.8	13.3	12.0		13.2	14.4	
In percent of exports	...	...	39.4			56.3	62.9	51.9	39.3	32.9	32.1		51.1	96.7	
PV of PPG external debt	**...**	**...**	**14.5**			**18.3**	**22.7**	**19.0**	**15.8**	**13.3**	**12.0**		**13.2**	**14.4**	
In percent of exports	**...**	**...**	**39.4**			**56.3**	**62.9**	**51.9**	**39.3**	**32.9**	**32.1**		**51.1**	**96.7**	
In percent of government revenues	**...**	**...**	**69.4**			**102.1**	**117.0**	**101.8**	**76.2**	**62.6**	**51.7**		**69.0**	**78.8**	
Debt service-to-exports ratio (in percent)	**1.5**	**2.2**	**3.2**			**6.8**	**9.2**	**10.5**	**7.3**	**7.6**	**7.6**		**3.9**	**7.6**	
PPG debt service-to-exports ratio (in percent)	**1.5**	**2.2**	**3.2**			**6.8**	**9.2**	**10.5**	**7.3**	**7.6**	**7.6**		**3.9**	**7.6**	
PPG debt service-to-revenue ratio (in percent)	**3.1**	**3.9**	**5.7**			**12.4**	**17.1**	**20.5**	**14.2**	**14.5**	**12.3**		**5.3**	**6.2**	
Total gross financing need (Billions of U.S. dollars)	0.5	0.2	0.6			1.0	0.5	0.7	0.5	0.6	0.9		1.3	3.1	
Non-interest current account deficit that stabilizes debt ratio	8.1	5.0	7.6			6.5	2.7	9.8	8.4	6.0	5.2		4.2	5.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	13.5	0.1	8.9	9.0	10.0	3.9	9.6	6.7	9.7	6.9	3.1	6.7	2.6	3.1	2.9
GDP deflator in US dollar terms (change in percent)	1.4	13.9	-2.6	10.2	11.9	0.0	7.7	3.9	3.7	2.0	0.4	3.0	1.8	2.0	1.8
Effective interest rate (percent) 5/	1.0	2.2	3.4	1.4	0.8	3.3	4.1	3.7	3.3	2.9	2.4	3.3	2.3	2.0	2.2
Growth of exports of G&S (US dollar terms, in percent)	23.3	22.3	-4.1	49.6	84.7	-8.4	31.5	12.4	25.1	9.9	-4.6	11.0	-3.9	0.2	-1.5
Growth of imports of G&S (US dollar terms, in percent)	18.1	11.3	3.8	13.0	23.8	-6.3	27.2	7.3	14.9	5.1	-2.6	7.6	-1.2	2.2	0.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	3.4	0.4	36.8	35.9	35.8	33.0	24.2	34.1	35.7	34.7
Government revenues (excluding grants, in percent of GDP)	18.9	23.2	20.9			17.9	19.4	18.6	20.7	21.3	23.1		19.2	18.3	19.0
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.5			0.4	0.4	0.5	0.6	0.6	0.7		0.9	1.6	
of which: Grants	0.1	0.2	0.3			0.3	0.4	0.4	0.4	0.4	0.4		0.6	1.1	
of which: Concessional loans	0.1	0.1	0.1			0.1	0.1	0.2	0.2	0.2	0.2		0.3	0.5	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			2.4	2.3	2.8	2.6	2.6	2.6		2.9	3.1	2.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			32.4	19.4	68.8	68.4	68.3	66.9		70.1	71.1	70.4
Memorandum items:															
Nominal GDP (Billions of US dollars)	10.7	12.2	12.9			13.4	15.8	17.6	20.0	21.8	22.6		27.7	45.2	
Nominal dollar GDP growth	15.1	14.0	6.0			4.0	18.1	10.9	13.8	9.1	3.5	9.9	4.4	5.2	4.7
PV of PPG external debt (in Billions of US dollars)			1.9			2.5	3.6	3.4	3.2	2.9	2.7		3.7	6.6	
(PVt-PVt-1)/GDPt-1 (in percent)						4.8	8.2	-1.7	-1.0	-1.3	-0.9	1.3	0.9	0.8	0.8
Gross workers' remittances (Billions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	14.5			18.3	22.7	19.0	15.8	13.3	12.0		13.2	14.4	
PV of PPG external debt (in percent of exports + remittances)	...	...	39.4			56.3	62.9	51.9	39.3	32.9	32.1		51.1	96.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	3.2			6.8	9.2	10.5	7.3	7.6	7.6		3.9	7.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	18	23	19	16	13	12	**13**	14
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	18	20	13	9	4	-1	**-15**	-33
A2. New public sector loans on less favorable terms in 2013-2033 2	18	24	20	17	15	14	**18**	24
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	18	25	23	19	16	14	**16**	17
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	18	31	39	34	31	30	**30**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	18	25	22	18	16	14	**15**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	18	30	32	28	25	24	**24**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	18	36	45	39	35	34	**34**	24
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	18	32	26	22	19	17	**18**	20
PV of debt-to-exports ratio								
Baseline	56	63	52	39	33	32	**51**	97
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	56	57	37	21	10	-2	**-58**	-222
A2. New public sector loans on less favorable terms in 2013-2033 2	56	66	55	43	37	38	**71**	164
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	56	63	52	40	33	32	**51**	97
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	56	125	190	151	135	141	**203**	253
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	56	63	52	40	33	32	**51**	97
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	56	85	88	70	62	63	**93**	127
B5. Combination of B1-B4 using one-half standard deviation shocks	56	120	158	125	113	117	**168**	208
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	56	63	52	40	33	32	**51**	97
PV of debt-to-revenue ratio								
Baseline	102	117	102	76	63	52	**69**	79
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	102	105	72	41	18	-3	**-79**	-181
A2. New public sector loans on less favorable terms in 2013-2033 2	102	122	108	83	71	62	**96**	134
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	102	131	122	92	75	62	**83**	94
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	102	162	211	166	146	129	**155**	117
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	102	129	119	89	73	60	**80**	92
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	102	157	173	135	117	102	**125**	103
B5. Combination of B1-B4 using one-half standard deviation shocks	102	187	239	189	166	147	**176**	131
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	102	163	142	106	87	72	**96**	110

Table 3b.Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	7	9	10	7	8	8	**4**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	7	9	9	6	6	5	**-1**	-9
A2. New public sector loans on less favorable terms in 2013-2033 2	7	9	12	10	10	5	**4**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	7	9	10	7	8	8	**4**	8
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	7	13	20	16	16	16	**17**	25
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	7	9	10	7	8	8	**4**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	7	9	11	8	9	9	**8**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	7	12	16	13	13	13	**14**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	7	9	10	7	8	8	**4**	8
Debt service-to-revenue ratio								
Baseline	12	17	21	14	14	12	**5**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	12	17	18	12	11	7	**-1**	-7
A2. New public sector loans on less favorable terms in 2013-2033 2	12	17	23	20	20	7	**5**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	12	19	25	17	17	15	**6**	7
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	12	17	22	17	17	15	**13**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	12	19	24	16	17	14	**6**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	12	17	22	16	16	14	**10**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	12	19	25	19	19	17	**15**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	12	24	28	20	20	17	**7**	9
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	23	23	23	23	23	23	**23**	23

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.